FEDERAL SPECIFICATION

BUCKRAM; COTTON

This specification was approved by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government.

1. CLASSIFICATION

1.1 Types and classes.—Buckram covered by this specification shall be of the following types and classes, as specified:

Type I.—For bookbinding.

Class 1.—4.7 oz. per square yard—light.

Class 2.—6.0 oz. per square yard—medium.

Class 3.—7.9 oz. per square yard—heavy.

Type II.—For clothing.

Class 1.—3.8 oz. per square yard.

Class 2.—4.5 oz. per square yard.

Class 3.—12.0 oz. per square yard.

Class 4.—17.0 oz. per square yard.

Type III.—For clothing, double texture.

Class 1.—32 to 34 oz. per square yard.

1.2 Color.—The color of the buckram shall be as specified.

2. APPLICABLE SPECIFICATIONS

2.1 Specifications. — The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

CCC–T–191—Textiles; General Specifications, Test Methods.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.—The fabric shall be of cotton, thoroughly cleaned and free from waste.

3.2 Weave.—The weave shall be plain except that type I, classes 2 and 3, shall be plain with 2 ends woven as one, and type II, class 3, shall be 4-harness herringbone twill, 2 up and 2 down, 20 ends right and 20 ends left.

3.3 Type I, for bookbinding[1].

3.3.1 *Physical requirements.* — Type I buckram shall conform to the requirements in table I.

3.3.2 *Finish.*—The buckram shall have a regular commercial finish and shall be free from pinholes, blemishes, and tackiness.

3.3.3 *Sizing.*—Unless otherwise specified, the buckram shall be starch-filled.

3.3.4 *Length.*—Unless otherwise specified, the buckram shall be furnished in rolls of one continuous piece not less than 40 yards long.

3.4 Type II, for clothing.

3.4.1 *Physical requirements.* — Type II buckram shall conform to the requirements in table II.

[1] The requirements comply with those in Book Cloths, Buckrams, and Impregnated Fabrics for Bookbinding Purposes except Library Bindings, Commercial Standard CS57–40.

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TABLE I.—*Physical requirements for type I*

Class	Number of yarns per inch, minimum			Breaking strength, minimum			Weight per sq. yd. stripped,[1] minimum	Wid minim
	Warp	Filling	Warp plus filling	Warp	Filling	Warp plus filling		
				Pounds	*Pounds*	*Pounds*	*Ounces*	*Inche*
1	30	25	58	65	35	110	4.7	38
2	83	21	106	100	55	165	6.0	38
3	83	24	110	120	70	200	7.9	38

[1] The basis of weight comparison shall be the weight of the fabric which remains after the finished fabric has been stripped of all starch and loading, or impregnating material. (See 4.4.)

TABLE II.—*Physical requirements for type II*

Class	Number of yarns per inch, minimum		Breaking strength, minimum		Finished weight per sq. yd., minimum	Sizing and other nonfibrous materials, maximum	Width, minimum
	Warp	Filling	Warp	Filling			
			Pounds	*Pounds*	*Ounces*	*Percent.*	*Inches*
1	46	44	50	40	3.8	35	35½
2	50	44	60	50	4.5	60	32
3	64	20	150	80	12.0	50	34
4	74	21	130	40	17.0	65	34

3.4.2 *Finish.*

3.4.2.1 *Class 1.*—The buckram shall have a "swing" finish and shall be free from tackiness and other objectionable features.

3.4.2.2 *Class 2.*—The buckram shall have a smooth, clear, permanent finish suitable for lining shirt collars. Finishes showing a loss in breaking strength of more than 10 percent when tested in accordance with 4.4, will not be acceptable.

3.4.2.3 *Classes 3 and 4.*—The buckram shall have a regular commercial finish and shall be free from tackiness and other objectionable features detracting from the original hard "finish".

3.4.3 *Shrinkage.*—Class 2 buckram shall not show a shrinkage exceeding 1 percent in either the warp or filling direction.

3.5 Type III, for clothing, double texture.—Type III buckram shall be made of two plies of type II, class 3, buckram and shall be jointed by sizing the two fabrics together. There shall be no objectionable odor caused by the sizing. The sizing shall in no way tender the buckram either at the time of manufacture or in storage. The weight of the buckram shall be not less than 32 ounces nor more than 34 ounces per square yard.

3.6 Workmanship.—The buckram shall be clean, evenly woven, and free from defects which affect its appearance or which may impair its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—A piece of buckram, at least 1½ yards long and the full width of the cloth, of each type, class, and color, shall be taken at random by the purchaser from each 1,000 yards or fraction thereof offered for delivery, except that when the shipment is

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over 10,000 yards, one piece shall be taken from each one-tenth of the shipment. The pieces shall be delivered to the laboratory without creases.

4.2 Inspection.—Inspection shall be made at the point of delivery or at the factory at the discretion of the purchasing agent.

4.3 Tests.—Tests shall be conducted in accordance with the applicable methods described in Federal Specification CCC–T–191, and as hereinafter specified.

4.3.1 *Breaking strength.*

4.3.1.1 *Type I, classes 1, 2, and 3, and type II, class 4,* shall be tested for breaking strength in accordance with the cut-strip method described in Federal Specification CCC–T–191, except that the specimens shall be obtained as follows:

A strip of buckram 7 inches long shall be cut from across the entire width of the piece. Twelve specimens shall be cut from this strip, six with the long dimension in the direction of the warp and six with the long dimension in the direction of the filling. Each specimen shall be 1 inch in width. (See fig. 1.)

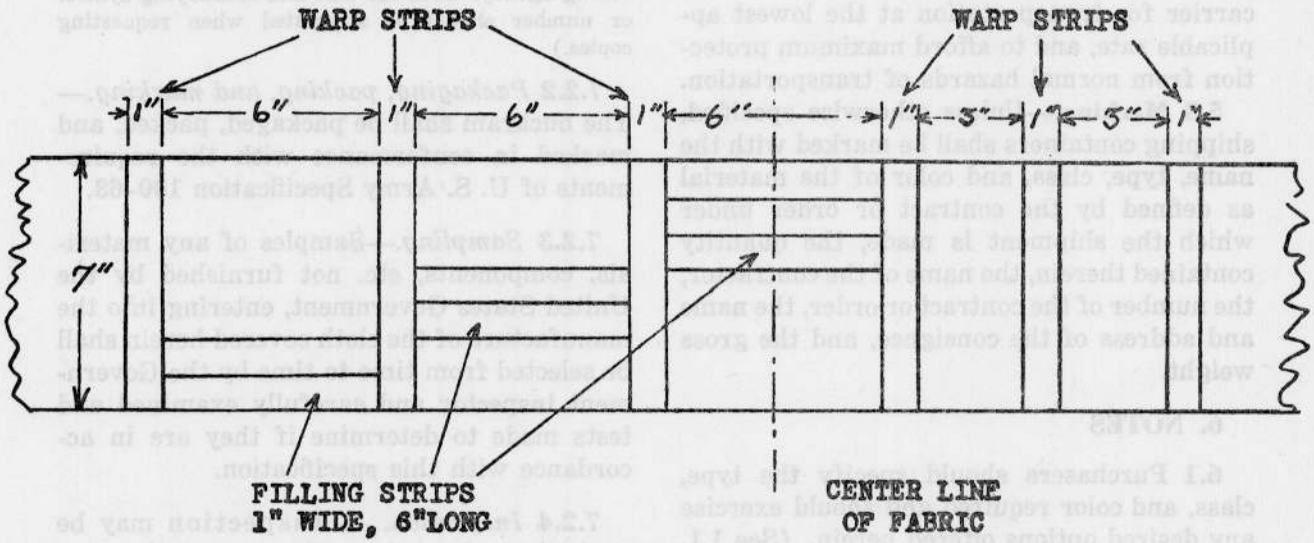


FIGURE 1.—Diagram for cutting test specimens for breaking strength test.

No two specimens shall contain the same yarns and care should be taken to insure that the yarns to be subjected to test have not been damaged during the cutting of the specimens.

4.3.1.2 *Type II, classes 1, 2, and 3,* shall be tested for breaking strength in accordance with the grab method described in Federal Specification CCC–T–191.

4.3.2 *Finish for type II, class 2.*—A piece of buckram, not less than 20 inches long, and full width, shall be treated for 1 hour with 0.7 percent available chlorine solution. The piece shall then be rinsed in a sour-solution and washed thoroughly until a control piece shows no trace of chlorine. The piece shall be air-dried and then conditioned for 4 hours in an atmosphere having a relative humidity of 65 ± 2 percent and a temperature of 75°± 5°F. Eight pairs of specimens, 4 warpwise and 4 fillingwise, with each pair having the same yarns, shall be taken for tests. Four warp specimens and four filling specimens shall be tested in a pendulum type of machine in accordance with the grab method described in Federal Specification CCC–T–191. The remaining warp and filling specimens shall be oven-dried at 105° to 110°C. for 2 hours and tested in the same manner as the first 8 specimens. The differences in the average breaking strength

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of the respective warp and filling samples shall be computed and the percentage of loss determined. If a specimen slips or breaks in or at the edge of the jaws, the result shall be discarded and another pair of specimens shall be taken, both having identical warp or filling yarns, as the case may be. The test shall be repeated on this pair of specimens.

5. PREPARATION FOR DELIVERY

5.1 Packaging. — Unless otherwise specified, commercial packages will be acceptable.

5.2 Packing.—The buckram shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.—Unless otherwise specified, shipping containers shall be marked with the name, type, class, and color of the material as defined by the contract or order under which the shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Purchasers should specify the type, class, and color required and should exercise any desired options offered herein. (See 1.1, 1.2, 3.3.3, 3.3.4, 4.2, 5.1, and 5.3.)

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following departmental specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army.

7.2.1 *Applicable specification.*

U. S. Army Specification:

100–63 — Packaging and Packing of Cotton Fabric (except Cotton Duck and Coated Cotton Fabrics).

(Copies of specifications required for Army purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.)

7.2.2 *Packaging, packing, and marking.*— The buckram shall be packaged, packed, and marked in conformance with the requirements of U. S. Army Specification 100–63.

7.2.3 *Sampling.*—Samples of any materials, components, etc. not furnished by the United States Government, entering into the manufacture of the cloth covered herein shall be selected from time to time by the Government inspector and carefully examined and tests made to determine if they are in accordance with this specification.

7.2.4 *Inspection.* — Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or material shall not relieve the contractor of responsibility for faulty workmanship or material which may be discovered at any time prior to final acceptance. Final inspection of the finished material shall be made either at point of production or at point of delivery designated in the contract or purchase order of the procuring agency. In case of factory inspection, every facility shall be afforded inspectors by the manufacturer, for the prosecution of their work.

7.3 Navy.

7.3.1 *Applicable specifications and other publications.*

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Navy Department Specification:

General Specifications for Inspection of Material.

Bureau of Supplies and Accounts Publication:

Navy Handbook on Packing and Materials Handling—Packaging, Packing, and Marking of Clothing, Small Stores and Textiles.

(Copies of Federal specifications required for Navy purchases, and Navy Department specifications may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.

(Applicable sections of the Navy Handbook on Packing and Materials Handling—Packaging, Packing, and Marking of Clothing, Small Stores and Textiles may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C. When requesting copies, bidders and contractors should state the specification number and complete nomenclature of the item to be packed, so that the proper sections may be forwarded.)

7.3.2 *Packaging, packing, and marking for shipment.* — The commodity shall be packed in accordance with the Navy Handbook on Packing and Materials Handling—Packaging, Packing, and Marking of Clothing, Small Stores and Textiles.

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